



A two-channel perfect reconstruction filter bank associated with the linear canonical transform

Shuai-Shuai Fan, Bing-Zhao Li & Tian-Zhou Xu

To cite this article: Shuai-Shuai Fan, Bing-Zhao Li & Tian-Zhou Xu (2014) A two-channel perfect reconstruction filter bank associated with the linear canonical transform, International Journal of Electronics Letters, 2:2, 72-82, DOI: [10.1080/00207217.2014.880954](https://doi.org/10.1080/00207217.2014.880954)

To link to this article: <https://doi.org/10.1080/00207217.2014.880954>



Published online: 06 Feb 2014.



Submit your article to this journal [↗](#)



Article views: 58



View related articles [↗](#)



View Crossmark data [↗](#)

A two-channel perfect reconstruction filter bank associated with the linear canonical transform

Shuai-Shuai Fan, Bing-Zhao Li* and Tian-Zhou Xu

School of Mathematics, Beijing Institute of Technology, Beijing 100081, China

(Received 15 February 2013; accepted 1 January 2014)

Sampling rate conversion is one of the most important concepts in modern signal processing community; it can be realised by interpolation and decimation operations. Based on the scaling operation and following the idea of the classical results, this paper investigate the sampling rate conversion associate with the discrete time linear canonical transform (DTLCT). First, we derive a new structure of sampling rate conversion for any rational or irrational factors in the linear canonical transform domain, then a two-channel nonuniform perfect reconstruction filter bank in the linear canonical transform domain is proposed based on the sampling rate conversion; the simulations are also performed to verify the derived results.

Keywords: linear canonical transform (LCT); scaling operation; sampling rate conversion; perfect reconstruction (PR); two-channel filter bank

1. Introduction

Sampling rate conversion is one of the hottest research topics in modern signal processing community. In traditional filter bank (FB) system, we also apply this method by performing interpolation and decimation with rational factors to design the perfect reconstruction FBs. Most of the published paper about the sampling rate conversion are about the rational factors and there few paper published about the irrational factors. Although irrational factors can be approximated by rational factors to reasonable accuracy, i.e., $r = L/M$, L and M might have to be very large with higher sampling rates. It is still interesting and worthwhile to investigate the sampling rate conversion of any rational and irrational factors, and the associated perfect reconstruction FBs.

Following this topic, many methods have been proposed about the sampling rate conversion based on the scaling operations associated with the classical Fourier transform (Lee, Zhao, Narasimha, & Rao, 2003; Pei & Kao, 2005; Ramstad, 1984; Xu & Li, 2013). A practical method has been proposed in Lee et al. (2003) to accomplish it using the concept of frequency domain warping. In contrast to the methods of Ramstad (1984), Xu and Li (2013), and Lee et al. (2003), Pei in Pei and Kao (2005) proposed a method to achieve the scaling operations, and his method can be realised with only one step to perform the sampling rate conversion. Based on this method, a new structure of a two-channel nonuniform perfect reconstruction filter bank (PRFB) in the Fourier transform domain has also been studied in detail (Pei & Kao, 2005). However, in real applications, we often encounter the non stationary signal, and they are often non-bandlimited in the traditional Fourier domain. If we still apply these methods to non-bandlimited signals, we

*Corresponding author. Email: li_bingzhao@bit.edu.cn

will obtain wrong or suboptimal results. Therefore, it is worthwhile to look for the other kind of transforms or domains to perform this actions.

At the same time, the linear canonical transform (LCT) (Ozaktas, Kutay, & Zalevsky, 2000; Tao, Deng, & Wang, 2009; Wolf, 1979) is shown to be one of the most powerful tools in signal processing community, it is an integral transform with three free parameters. The non-bandlimited signals of the Fourier transform domain can be bandlimited in the LCT domain (Li, Tao, & Wang, 2007), and many transforms such as the Fourier transform, the fractional transform (FrFT), the Fresnel transform and scaling operations are special cases of the LCT. It has found many applications in radar system analysis, filter design, phase retrieval, pattern recognition and many other areas (Barshan, Kutay, & Ozaktas, 1997; Li, Li, & Xu, 2012; Meng, Tao, & Wang, 2007; Sharma & Joshi, 2006; Wei, Ran, & Li, 2011; Wei, Ran, Li, & Ma, 2009; Yan, Li, Chen, & Cheng, 2013; Zhao, Tao, & Wang, 2008). The classical sampling rate conversion concepts (interpolation and decimation) associated with the LCT domain was introduced in Zhao et al. (2008), moreover, they also derive the formulae of sampling rate conversion by real factors for the LCT in time domain, and the results can be looked as the generalisation of the classical results in the Fourier domain to the LCT domain.

Following the ideas of the classical method in Pei and Kao (2005), we apply the scaling operations to investigate the sampling rate conversion problem and the associated PRFB in the LCT domain in this paper in detail. The paper is organized as follows, the preliminary is given in Section 2, the main results of the irrational sampling rate conversion and the associated PRFB in Section 3, simulations are also given in Section 4 to verify the achieved results and finally, we make conclusions in Section 5.

2. Preliminary

2.1. The linear canonical transform (LCT)

The LCT of a signal $x(t)$ with parameter A is defined as follows (Xu & Li, 2013).

$$X_{a,b,c,d}(u) = L_A[x(t)](u) = \begin{cases} \int_{-\infty}^{\infty} x(t) \sqrt{\frac{1}{j2\pi b}} e^{j\left(\frac{a}{2}t^2 - \frac{2}{b}ut + \frac{d}{2}u^2\right)} dt, & b \neq 0 \\ \sqrt{d} e^{j\frac{c}{2}du^2} x(du), & b = 0 \end{cases} \quad (1)$$

where $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the parameter matrix of LCT satisfying $ad - bc = 1$, i.e., $\det(A) = 1$.

The inverse transform of the LCT (ILCT) is given by the LCT having parameter $A^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Hence, the original signal $x(t)$ can be derived from $L_A[x](u)$ via

$$\begin{aligned} x(t) &= L_{A^{-1}}[L_A[x](u)](t) \\ &= \sqrt{\frac{1}{j2\pi b}} e^{(-ja/2b)t^2} \int_{-\infty}^{+\infty} L_A[x](u) e^{-j(d/2b)u^2} \times e^{j(1/b)ut} du \end{aligned} \quad (2)$$

With the rapid development of the modern signal processing methods and the achievements of the fractional Fourier transform, the LCT is shown to be one of the most important tools for signal processing and optical analysis. The sampling theories have been studied extensively in recently years in Tao, Li, and Wang (2008), Healy and Sheridan (2009), Li and Zhu (2012a, 2012b), We, Ran, and Li (2013) and Xiao and

Sun (2013); the other concepts such as the convolution theorem (Deng, Tao, & Wang, 2006; We, Ran, & Li, 2012), the uncertainty principle and the eigenfunctions (Ding & Pei, 2013; Pei & Ding, 2012; Zhao, Tao, Li, & Wang, 2009) are also investigated in the LCT domain. The digital computational methods and the applications of the LCT in filtering, time-frequency analysis and in speech signal processing are proposed in Koc, Ozaktas, Candan, and Kutay (2008), Oktem and Ozaktas (2009), Bai, Li, and Cheng (2012), Qiu, Li, and Li (2013) and Kou, Morais, and Zhang (2013).

2.2. Sampling rate conversion

The sampling rate conversion is one of the most important concepts in the multirate signal processing community. The interpolation and the decimation are two basic operations in sampling rate conversion system. The sampling rate conversion also plays an important role in multichannel FBs, the analysis FBs and the synthesis FBs can be realised by the careful design of the PRFFB in real situations. For more results about the sampling rate conversion and the associated FBs, one can refer to Vaidyanathan (1992) and Fliege (2000).

If we assume that the original signal $x(t)$ is band-limited in the LCT domain, i.e., $X_{(a,b,c,d)}(u) = 0$ for $|u| > u_c$, then it is shown in Zhao et al. (2008) that the classical operations of interpolation and decimation can be generalised to the LCT domain, considering the discrete time signal $x[n]$ is sampled from the continuous signal $x(t)$ with the sampling period Δt_x . To increase the sampling rate of the original discrete signal $x[n]$ by an integer factor L , the output sequence $y[n]$ is

$$y[n] = \begin{cases} x[n/L] & n = Lk, k \in \mathbb{Z}, \\ 0 & \text{otherwise.} \end{cases}$$

The sampling period of $y[n]$ is reduced to $\Delta t_y = \Delta t_x/L$, and the LCT with parameter A of $y[n]$ is

$$Y_{a,b,c,d}(\omega) = X_{(a,b,c,d)}(L\omega) \quad (3)$$

Otherwise, reduce the sampling rate of the origin discrete signal $x[n]$ by an integer factor D , then the output sequence $y[n]$ is

$$y[n] = x[Dn]$$

and the sampling period of $y[n]$ is reduced to $\Delta t_y = D\Delta t_x$, the LCT of $y[n]$ with parameter A is

$$Y_{a,b,c,d}(\omega) = \frac{1}{D} \sum_{k=0}^{D-1} X_{(a,b,c,d)}\left(\frac{\omega - 2\pi k}{D}\right) e^{j2\pi k b d \frac{\text{sgn}(b)\omega - \pi k}{(D\Delta t_x)^2}} \quad (4)$$

According to the results derived in Zhao et al. (2008), we usually first increase the sampling rate of the origin signal, and pass a low-pass filter and then decrease the sampling rate to perform the sampling rate conversion associated with the LCT. Unlike this method, we propose a new kind of sampling rate conversion model based on the scaling operation for any rational and irrational factors in this paper.

3. Main results

In this section, we first investigate the scaling operations associated with the LCT, and give a different proof the results in Zhao et al. (2008) for sampling rate conversion for any real factors based on the scaling operation, and then a new two-channel PRFB is proposed for any rational or irrational factors.

3.1. Scaling operation in the LCT domain

In fact, the scaling operation can be used to perform the up-sampling and down-sampling (Pei & Kao, 2005), suppose the scaled version of a discrete time signal $x[n]$ is $x_\beta[n]$, where β is the scaling factor and $\beta \in R, \beta > 0$. In this way, $\beta > 1$ is equivalent to interpolation, and $0 < \beta < 1$ is for decimation. The following results can be derived in the LCT domain associated with the scaling operation.

Theorem 1. To change the sampling rate of $x[n]$ by real factor $\beta > 1$, it can be achieved by the following formula

$$x_\beta[n] = e^{-j(a/2b)n^2(\frac{\Delta t_x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2 \Delta t_x^2} \text{sinc}\left(\frac{n}{\beta} - k\right) \quad (5)$$

Proof. The scaling property of a continuous time LCT (CTLCT) of a signal $x(t)$ can be derived as following.

$$\begin{aligned} x(t) &\leftrightarrow X_{(a,b,c,d)}(u) \\ x\left(\frac{t}{\beta}\right) &\leftrightarrow \beta X_{(a_1,b_1,c_1,d_1)}(\beta u) \end{aligned} \quad (6)$$

where

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} = \begin{pmatrix} \beta^2 a & b \\ c & \frac{d}{\beta^2} \end{pmatrix}$$

The discrete time LCT (DTLCT) of $x[n]$ can be written as (Zhao et al., 2008)

$$X_{(a,b,c,d)}(\omega) = \sqrt{\frac{1}{j2\pi b}} e^{j(d/2b)(\frac{b\omega}{\Delta t})^2} \sum_{n=-\infty}^{+\infty} x[n] e^{j(a/2b)n^2 \Delta t_x^2} \quad (7)$$

and the corresponding inverse transform is

$$x[n] = \sqrt{\frac{j\bar{b}}{2\pi}} e^{j(a/2b)n^2(\Delta t_x)^2} \int_{-\pi}^{\pi} X_{(a,b,c,d)}(w) e^{-j(d/2b)(b\omega/\Delta t_x)^2} e^{jn\omega} d\omega$$

we could reasonably derive the DTLCT counterpart of Equation (6), i.e.,

$$\begin{aligned} x[n] &\leftrightarrow X_{(a,b,c,d)}(\omega) \\ x_\beta[n] &\leftrightarrow \beta X_{(a_1,b_1,c_1,d_1)}(\beta\omega) \end{aligned} \quad (8)$$

From the ILCT, we obtain

$$\begin{aligned} x_\beta[n] &= \text{IDTLCT}(\beta X_{(a_1,b_1,c_1,d_1)}(\beta\omega)) \\ &= \frac{\beta}{2\pi} e^{-j(a/2b)n^2(\frac{\Delta t_x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a_1/2b_1)k^2\Delta t_x^2} \int_{-\frac{\pi}{\beta}}^{\frac{\pi}{\beta}} e^{j(n-k\beta)\omega} d\omega \end{aligned} \quad (9)$$

Due to interpolation, the new frequency period of $x_\beta[n]$ in the LCT domain is equal to $2\pi/\beta$; we need only preserve those frequency components inside $[-\pi/\beta, \pi/\beta]$ for processing, so we change the upper bound and lower bound of the definite integral of Equation (9) from $[-\pi, \pi]$ to $[-\pi/\beta, \pi/\beta]$.

Equation (9) can be divided into two disjoint cases, one being $k = n/\beta$ and the other being $k \neq n/\beta$. For $k = n/\beta$

$$\int_{-\frac{\pi}{\beta}}^{\frac{\pi}{\beta}} e^{j(n-k\beta)\omega} d\omega = \int_{-\frac{\pi}{\beta}}^{\frac{\pi}{\beta}} d\omega = \frac{2\pi}{\beta} \quad (10)$$

and for $k \neq n/\beta$

$$\int_{-\frac{\pi}{\beta}}^{\frac{\pi}{\beta}} e^{j(n-k\beta)\omega} d\omega = \frac{2\pi}{\beta} \text{sinc}\left(\frac{n}{\beta} - k\right) \quad (11)$$

From Equations (7) to (11), we obtain

$$\begin{aligned} x_\beta[n] &= \frac{\beta}{2\pi} e^{-j(a/2b)n^2(\frac{\Delta t_x}{\beta})^2} \left[\sum_{k=-\infty, k=n/\beta}^{\infty} x[k] e^{j(a_1/2b_1)k^2\Delta t_x^2} \int_{-\frac{\pi}{\beta}}^{\frac{\pi}{\beta}} e^{j(n-k\beta)\omega} d\omega \right. \\ &\quad \left. + \sum_{k=-\infty, k \neq n/\beta}^{\infty} x[k] e^{j(a_1/2b_1)k^2\Delta t_x^2} \int_{-\frac{\pi}{\beta}}^{\frac{\pi}{\beta}} e^{j(n-k\beta)\omega} d\omega \right] \\ &= e^{-j(a/2b)n^2(\frac{\Delta t_x}{\beta})^2} \left[\sum_{k=-\infty, k=n/\beta}^{\infty} x[k] e^{j(a_1/2b_1)k^2\Delta t_x^2} \right. \\ &\quad \left. + \sum_{k=-\infty, k \neq n/\beta}^{\infty} x[k] e^{j(a_1/2b_1)k^2\Delta t_x^2} \text{sinc}\left(\frac{n}{\beta} - k\right) \right] \\ &= e^{-j(a/2b)n^2(\frac{\Delta t_x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2\Delta t_x^2} \text{sinc}\left(\frac{n}{\beta} - k\right) \end{aligned} \quad (12)$$

□

The result of Theorem 1 is only suitable in the case where $\beta > 1$, i.e., interpolation. For the case where $0 < \beta < 1$, i.e., decimation, the upper and lower bound should be changed back to $[-\pi, \pi]$ again to avoid the aliasing due to down-sampling. This is equivalent to applying an

Table 1. Scaling formulas.

Interpolation	$\beta > 1$	$e^{-j(a/2b)n^2(\frac{\Delta x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2 \Delta x^2} \text{sinc}(\frac{n}{\beta} - k)$
Decimation	$\beta < 1$	$\beta e^{-j(a/2b)n^2(\frac{\Delta x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2 \Delta x^2} \text{sinc}(n - \beta k)$

ideal low-pass filter before down-sampling. Here, the new period of $X(\omega)$ due to decimation is larger than 2π , which forces us to abandon those frequency components outside 2π , or they will inevitably have aliasing images reflected into the basic 2π period and damage the algorithms. The following theorem can be derived for this kind of scaling operation.

Theorem 2. To change the sampling rate of signal $x[n]$ by real factor $\beta < 1$, it can be achieved by the following formula

$$x_\beta[n] = \beta e^{-j(a/2b)n^2(\frac{\Delta x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2 \Delta x^2} \text{sinc}(n - \beta k) \quad (13)$$

Proof. Similar with the proof of the Theorem 1, the signal $x_\beta[n]$ can be derived as

$$\begin{aligned} x_\beta[n] &= \text{IDTLCT}(\beta X_{(a_1, b_1, c_1, d_1)}(\beta \omega)) \\ &= \frac{\beta}{2\pi} e^{-j(a/2b)n^2(\frac{\Delta x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2 \Delta x^2} \int_{-\pi}^{\pi} e^{j(n-k\beta)\omega} d\omega \\ &= \beta e^{-j(a/2b)n^2(\frac{\Delta x}{\beta})^2} \sum_{k=-\infty}^{\infty} x[k] e^{j(a/2b)k^2 \Delta x^2} \text{sinc}(n - \beta k) \quad (14) \end{aligned}$$

□

For convenience, the main results about the interpolation and decimation operations associate with the LCT are summarised in Table 1. It can be seen that the derived results can be looked as the generalisation of classical results in Pei and Kao (2005) to the LCT domain.

3.2. Two-channel perfect reconstruction filter bank

Based on the derived results of scaling operation associated with the LCT, a new kind of two-channel PRFB is proposed in this subsection. The fundamental structure of the PRFB using irrational scaling factors associated with the LCT is shown in Figure 1, where $\beta, \gamma \in \mathbb{R}$, $\beta, \gamma > 1$ and $(1/\beta) + (1/\gamma) = 1$.

Note that there is no need to design the pre-filters and post-filters in the proposed PRFB structure since those filters have already been merged into the down-sampling and up-sampling kernels, respectively. The equivalent pre-filter addressed is the ideal low-pass filter whose cut-off frequencies are located at $\omega = \pm\pi$, and the equivalent post-filter is the

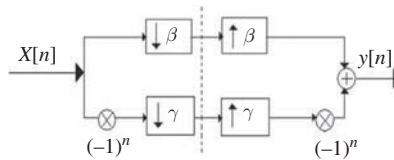


Figure 1. Irrational scaling two-channel PRFB structure.

ideal low-pass filter whose cut-off frequencies are located at $\omega = \pm(\pi/\beta)$. The basic goal of the proposed structure is to perfectly reconstruct the output such that $y[n] = x[n]$, under any circumstances, even if β and γ are irrational numbers.

4. Simulations

In order to verify the derived results, we take a 256-point normalised Gaussian random signal with zero mean and unit variance as input sequence $x[n]$, choosing $\beta = \sqrt{3}$ and $\gamma = (3 + \sqrt{3})/2$ to construct a two-channel nonuniform PRFB. The parameters of LCT is chosen as $(a, b, c, d) = (2, 1, 1, 1)$ in the simulation.

Figure 2 plots the samples of the original signal, following the results of the two-channel PRFB shown in Figure 1. The original signal can be reconstructed by two channels, the down sampled signals in channels 1 and 2 are plotted in Figures 3 and 4, respectively. The final reconstructed signal is shown in Figure 5, and the reconstruction errors are plotted in Figure 6. The simulation verifies the derived results of the paper.

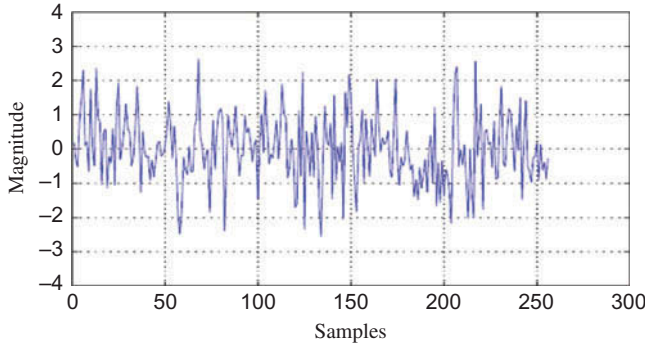


Figure 2. Original signal.

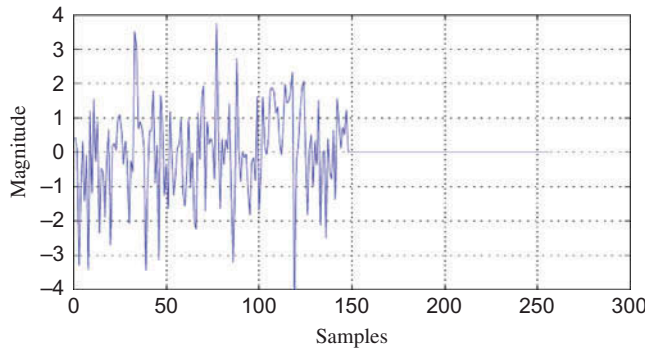


Figure 3. Down-sampled signals in channels 1.

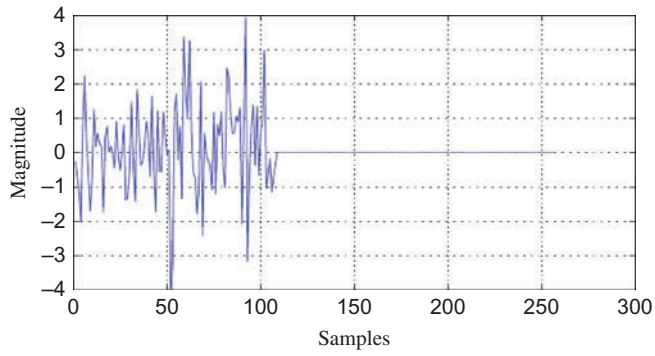


Figure 4. Down-sampled signals in channels 2.

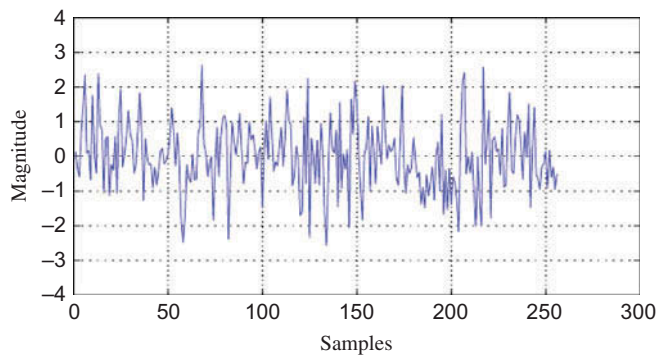


Figure 5. Reconstructed signal.

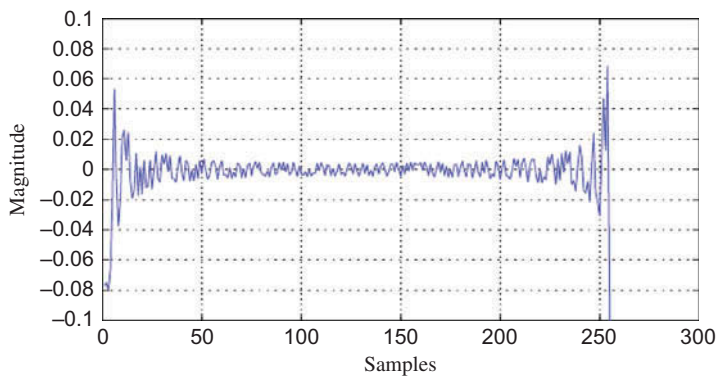


Figure 6. Reconstruction errors.

5. Discussions and conclusions

It should be noted that the results of Pei and Kao (2005) is more suitable for the bandlimited signal in Fourier domain, while the results of this paper may be more suitable for the non-bandlimited signal in Fourier domain (bandlimited signal in the LCT domain). The derived results in this paper can be looked as the generalisation of the classical results in Pei and Kao (2005) to the LCT domain. In order to show the comparative study with Pei and Kao (2005) through a graph to verify the derived result, we perform the simulation for the following signal

$$f(t) = e^{i(-1/2t^2 + 106t)}$$

Figure 7 plots the reconstruction errors using the Fourier domain method proposed in Pei and Kao (2005), while Figure 8 plots the reconstruction errors using the LCT domain method proposed in this paper. It is obvious to see that the method proposed in this paper is more suitable for this kind of signals, and the reconstruction errors are more littler as compared to the existing results. As compared to the results of Zhao et al. (2008), this paper derived the results based on the scaling operation, while Zhao et al. (2008) using the similar ideas of the classical method in Fourier domain.

Based on the scaling operation, this paper discussed the sampling rate conversion for any rational or irrational factors in the LCT domain. Following the derived results, a two-channel PRFB associated with the LCT is proposed. The simulations are also performed to verify the derived results. As compared to the existing results, the model proposed in this paper is more suitable for the non-bandlimited signal in the Fourier domain, and we can obtain the concise closed form formula in time domain from the results of the paper.

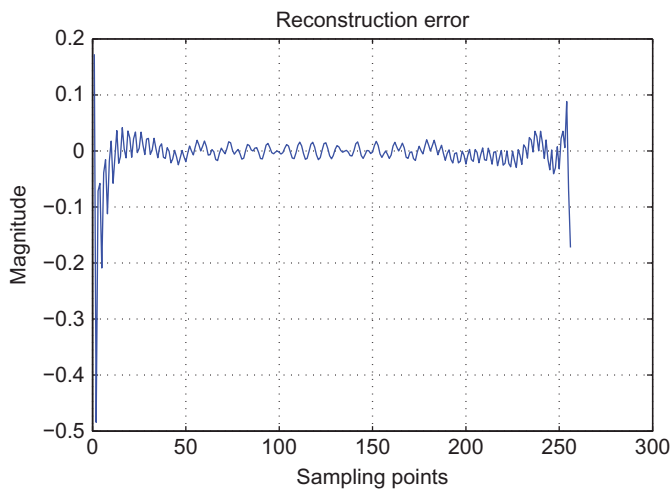


Figure 7. Reconstruction errors in Fourier domain.

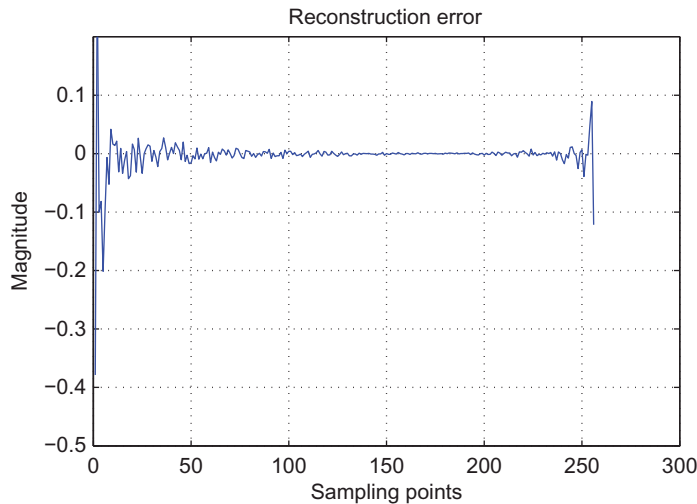


Figure 8. Reconstruction errors in LCT domain.

Acknowledgements

The authors thank the anonymous reviewers and the handing editor for their valuable comments and suggestions for the improvements of this manuscript. The authors also thank Prof. Huafei Sun of Beijing Institute of Technology for many fruitful discussions and the proofreading of this manuscript.

Funding

This work was supported by the National Natural Science Foundation of China [No. 61072097], [No. 61171195], and also supported by Program for New Century Excellent Talents in University [No. NCET-12-0042].

References

- Bai, R.-F., Li, B.-Z., & Cheng, Q.-Y. (2012). Wigner-Ville distribution associated with the linear canonical transform. *Journal of Applied Mathematics*, 2012, 1–11.
- Barshan, B., Kutay, M. A., & Ozaktas, H. M. (1997). Optimal filters with linear canonical transformations. *Optical Communications*, 135, 32–36.
- Deng, B., Tao, R., & Wang, Y. (2006). Convolution theorems for the linear canonical transform and their applications. *Science in China Series F: Information Sciences*, 49, 592–603.
- Ding, J.-J., & Pei, S.-C. (2013). Heisenberg's uncertainty principles for the 2-D nonseparable linear canonical transforms. *Signal Processing*, 93, 1027–1043.
- Fliege, N. J. (2000). *Multirate digital signal processing: Multirate systems-filter banks-wavelets*. New York: Wiley.
- Healy, J. J., & Sheridan, J. T. (2009). Sampling and discretization of the linear canonical transform. *Signal Processing*, 89, 641–648.
- Koc, A., Ozaktas, H. M., Candan, C., & Kutay, M. A. (2008). Digital computation of linear canonical transforms. *IEEE Transactions on Signal Processing*, 56, 2383–2394.
- Kou, K., Morais, J., & Zhang, Y. (2013). Generalized prolate spheroidal wave functions for offset linear canonical transform in Clifford analysis. *Mathematical Methods in the Applied Science*, 36, 1028–1041.
- Lee, S., Zhao, W., Narasimha, R., & Rao, R.-M. (2003). Discrete-time models for statistically self-similar signals. *IEEE Transactions on Signal Processing*, 51, 1221–1230.

- Li, B.-Z., Tao, R., & Wang, Y. (2007). New sampling formulate related to linear canonical transform. *Signal Processing*, 87, 983–990.
- Li, B.-Z., & Xu, T.-Z. (2012). Sampling in the linear canonical transform domain. *Mathematical Problems in Engineering*, 2012, 1–14.
- Li, B.-Z., & Xu, T.-Z. (2012). Spectral analysis of sampled signals in the linear canonical transform domain. *Mathematical Problems in Engineering*, 2012, 1–19.
- Li, C.-P., Li, B.-Z., & Xu, T.-Z. (2012). Approximating bandlimited signals associated with the LCT domain from nonuniform samples at unknown locations. *Signal Processing*, 92, 1658–1664.
- Meng, X.-Y., Tao, R., & Wang, Y. (2007). The fractional Fourier domain analysis of decimation and interpolation. *Science in China Series F: Information Sciences*, 50, 521–538.
- Oktem, F. S., & Ozaktas, H. M. (2009). Exact relation between continuous and discrete linear canonical transforms. *IEEE Transactions on Signal Processing*, 57, 727–730.
- Ozaktas, H. M., Kutay, M. A., & Zalevsky, Z. (2000). *The fractional fourier transform with applications in optics and signal processing*. New York: Wiley.
- Pei, S.-C., & Ding, J.-J. (2002). Eigenfunctions of linear canonical transform. *IEEE Transactions on Signal Processing*, 50, 11–26.
- Pei, S.-C., & Kao, M.-P. (2005). A two-channel nonuniform perfect reconstruction filter bank with irrational down-sampling factors. *IEEE Signal Processing Letters*, 12, 116–119.
- Qiu, W., Li, B.-Z., & Li, X.-W. (2013). Speech recovery based on the linear canonical transform. *Speech Communication*, 55, 40–50.
- Ramstad, T. A. (1984). Digital methods for conversion between arbitrary sampling frequencies. *IEEE Transactions on Acoustic, Speech, Signal Processing*, 30–32, 577–591.
- Sharma, K. K., & Joshi, S. D. (2006). Signal separation using linear canonical and fractional Fourier transforms. *Optical Communications*, 265, 454–460.
- Tao, R., Deng, B., & Wang, Y. (2009). *Theory and applications of the fractional fourier transform*. Beijing: Tsinghua University Press.
- Tao, R., Li, B.-Z., & Wang, Y. (2008). On sampling of bandlimited signals associated with the linear canonical transform. *IEEE Transactions on Signal Processing*, 56, 5454–5464.
- Vaidyanathan, P. P. (1992). *Multirate systems and filter banks*. Upper Saddle River, NJ: Prentice Hall.
- We, D., Ran, Q., & Li, Y. (2012). A convolution and correlation theorem for the cinear canonical transform and its application. *Circuits, Systems, and Signal Processing*, 31, 301–312.
- We, D., Ran, Q., & Li, Y. (2013). Sampling of bandlimited signals in the linear canonical transform domain. *Signal Image and Video Processing*, 7, 553–558.
- Wei, D., Ran, Q., & Li, Y. (2011). Reconstruction of band-limited signals from multichannel and periodic nonuniform samples in the linear canonical transform domain. *Optics Communications*, 284, 4307–4315.
- Wei, D., Ran, Q., Li, Y., & Ma, J. (2009). A convolution and product theorem for the linear canonical transform. *IEEE Signal Processing Letters*, 16, 853–856.
- Wolf, K. B. (1979). *Integral transform in science and engineering*. New York: Plenum Press.
- Xiao, L., & Sun, W. (2013). Sampling theorems for signals periodic in the linear canonical transform domain. *Optical Communications*, 290, 14–18.
- Xu, T.-Z., & Li, B.-Z. (2013). *Linear canonical transform and its applications*. Beijing: Science Press.
- Yan, J.-P., Li, B.-Z., Chen, Y.-H., & Cheng, Q.-Y. (2013). Wigner distribution moments associated with the linear canonical transform. *International Journal of Electronics*, 100, 473–481.
- Zhao, J., Tao, R., Li, Y.-L., & Wang, Y. (2009). Uncertainty principles for linear canonical transform. *IEEE Transactions on Signal Processing*, 57, 2856–2858.
- Zhao, J., Tao, R., & Wang, Y. (2008). Sampling rate conversion for linear canonical transform. *Signal Processing*, 88, 2825–2832.